

Workshop on Advanced Sampling Methodologies

Introduction to sampling design

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Workshop's agenda

- **Day 1:** Introduction to QGIS
- **Day 2:** Automatic update of census EA and national sampling frame
- **Day 3:** Introduction to R programming
- **Day 4: Introduction to sampling design**
- **Day 5:** Advanced R programming
- **Day 6:** Earth observation (EO) for sampling design
- **Day 7:** AI and machine learning for sample selection
- **Day 8:** Simulation and optimization of sampling strategies
- **Day 9:** R Shiny for sampling design evaluation
- **Day 10:** Wrap-up and workshop evaluation

Course materials

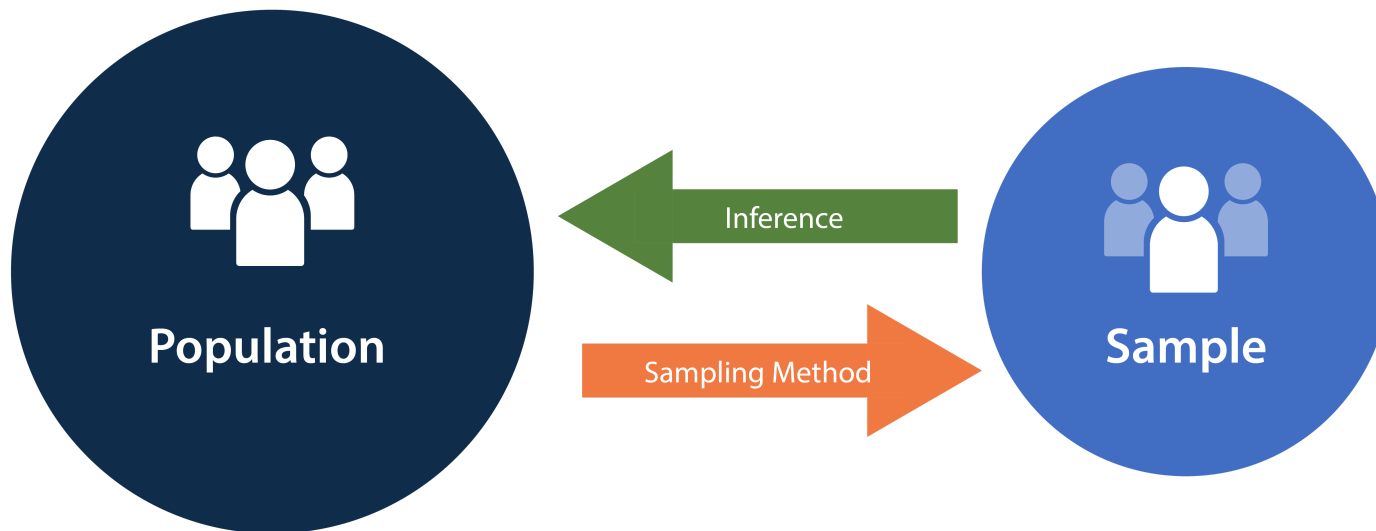
1. Today's course materials are on **Github**:
https://github.com/gianlucaboo/sampling_design_workshop
2. Click on **Day 4: Introduction to Sampling Design**
3. **Copy the url** and **paste it** on <https://download-directory.github.io>
4. Save the zipped directory on the workshop and unzip it

Today's agenda

- Introduction to sampling concepts
- Probability sampling methods
- Weighted sampling and unequal probability designs
- Practical implementation in R
- Design considerations and summary
- Practical exercises

Sampling

- **Definition:** sampling is the process of selecting a subset (called a **sample**) from a larger **population** to estimate characteristics of that population.
- **Purpose:** enables inference when measuring the entire population is costly or impractical. This offers the following advantages:
 - **Cost-effective:** sampling requires fewer resources than a full census.
 - **Feasibility:** some populations cannot be fully measured.
 - **Timely insights:** faster data collection for timely decision-making.



Key concepts of sampling (source: National Library of Medicine).

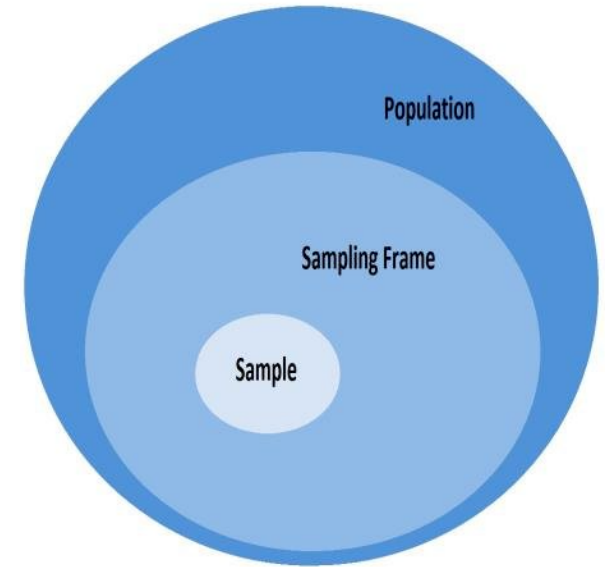
Key concepts

Population

- The **entire set of elements** (people, objects, events) you want to study.
- Provides the **scope** of inference for your study.
- **Example**: all university students in a country.

Sampling frame

- The **list or mechanism** used to identify and access members of the population.
- Ideally a sampling frame **should be**:
 - **Complete**: covers all members of population.
 - **Accurate**: no duplicates, no outdated entries.
 - **Accessible**: researcher can reach listed units.
- **Example**: an official student enrollment database.



Population, sampling frame, and sample (source: Ktistakis et al. 2021).

i If the sampling frame does not fully represent the population, **sampling bias** may occur.

Key concepts

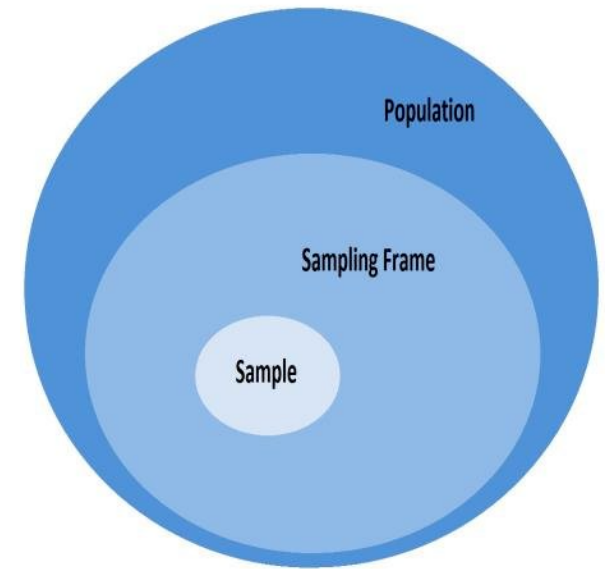
Sample

- A **subset of the population** selected for analysis.
- Used to make **inferences** about the population.
- **Example:** 1,000 students chosen from the national student population.

Sampling

- The **process or method** of selecting units from the population to form a sample.
- The **goal** is achieving a representative and unbiased sample.
- **Example:** students are selected randomly.

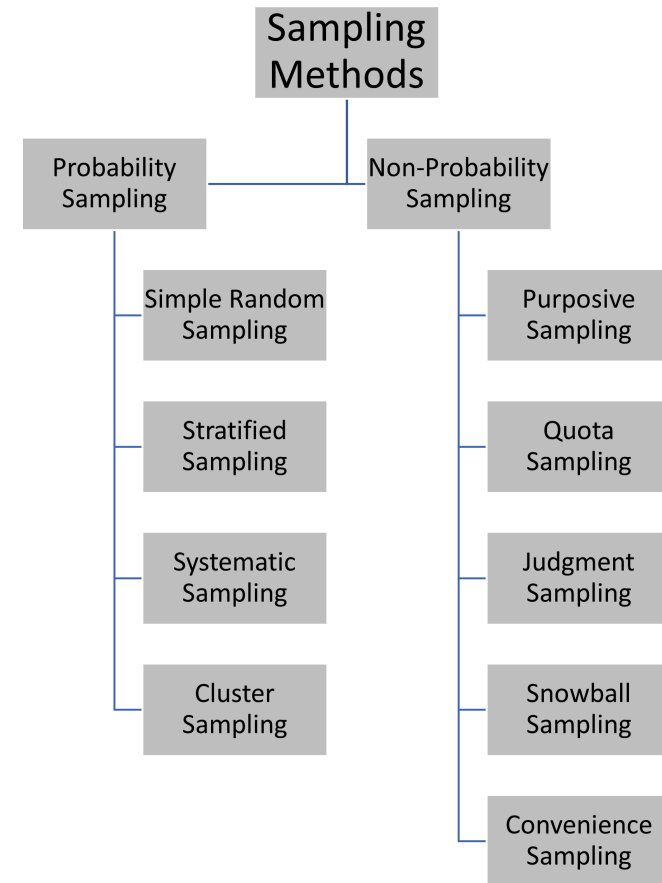
i **Sampling design** formalizes the plan for selecting a representative sample.



Population, sampling frame, and sample (source: Ktistakis et al. 2021).

Sampling design

- A **sampling design** defines how samples are chosen and the probability each unit has of being selected.
- **Sampling design** includes aspects like:
 - Probability versus non-probability sampling
 - Sampling technique
 - Sample size
 - Allocation strategies
 - Bias and representativeness

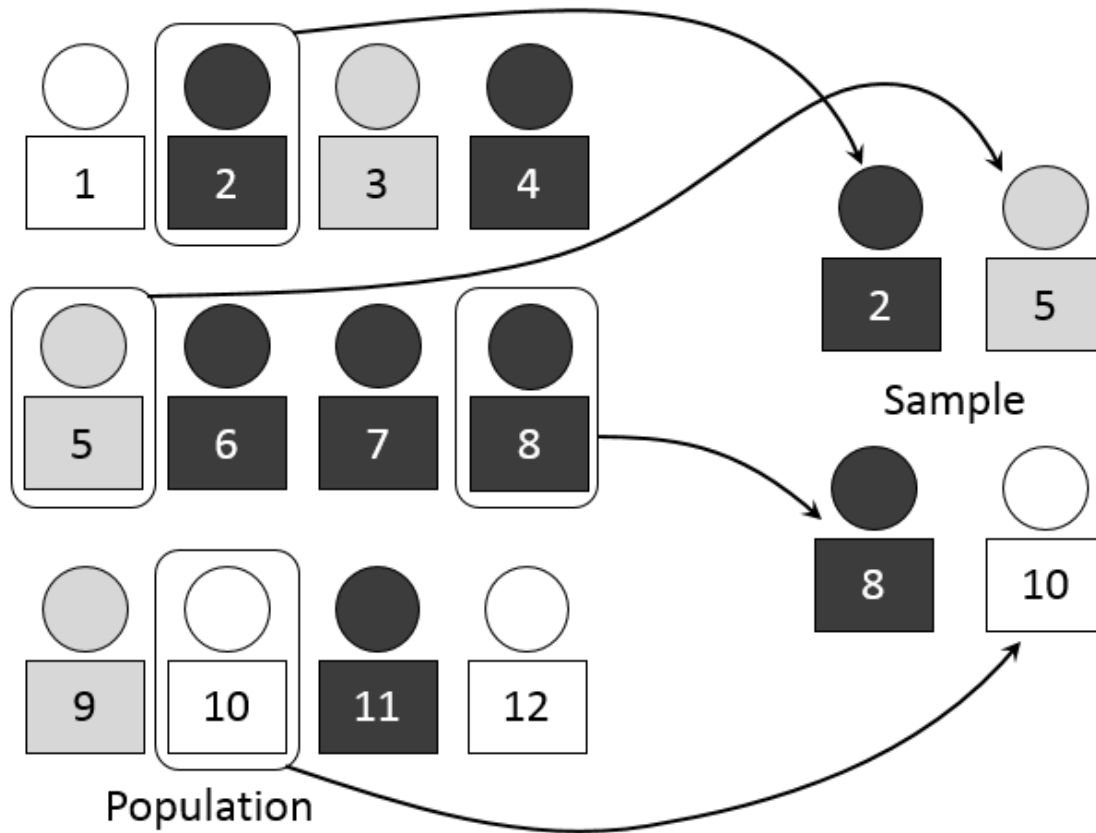


Sampling methods: probabilistic versus non-probabilistic.

 In this workshop we will cover **probability sampling** only.

Simple random sampling

- Every possible subset of size n from a population N has an **equal probability of being selected**.
- **Used when** complete list of the population is available.



Simple random sampling

Sample mean

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$$

x : sampled values

μ : population mean

σ^2 : population variance

n : sample size

N : population size

Sample variance

$$s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$$

Population mean

$$\mu = \frac{1}{N} \sum_{i=1}^N x_i$$

Population variance

$$\sigma^2 = \frac{1}{N} \sum_{i=1}^N (x_i - \mu)^2$$

 The sample mean can be computed by substituting N with n , while the sample variance by substituting N with n and $n - 1$ in the denominator (Bessel's correction).

Simple random sampling

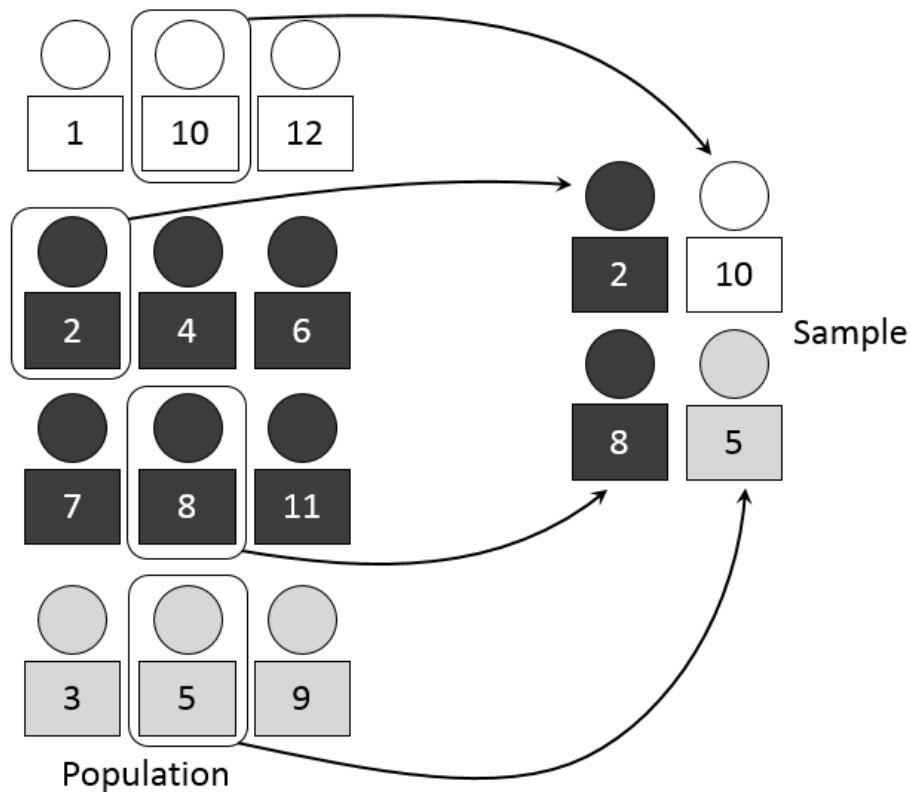
Implementation in R

```
1 library(tidyverse)           # Load tidyverse for data manipulation
2 set.seed(123)                # Set seed for reproducibility
3
4 N <- 10000                    # Define population size
5 pop <- data.frame(id = 1:N,   # Create population dataframe with IDs
6                   y = rlnorm(N, # Generate log-normal random values
7                           meanlog = 2,
8                           sdlog = 0.5))
9
10 n <- 500                     # Define sample size
11 s_ids <- sample(pop$id, size = n, # Randomly select n IDs without replacement
12               replace = FALSE)
13 samp <- pop |> filter(pop$id %in% s_ids) # Extract sampled observations
14
15 ybar_hat <- mean(samp$y)      # Sample mean of y
16 Y_hat <- N * ybar_hat        # Estimate of population total (N * ybar_hat)
```

 Try to run the code in your Script Editor.

Stratified sampling

- Population is divided into **homogeneous subgroups (strata)** and a **simple random sample** is drawn **within each stratum**.
- Improves efficiency by reducing variability **within strata**.



Stratified sampling

Sample mean

$$\bar{x}_{st} = \sum_{h=1}^H W_h \bar{x}_h$$

Approximate variance

$$\text{Var}(\bar{x}_{st}) = \sum_{h=1}^H W_h^2 \frac{\sigma_h^2}{n_h} \left(1 - \frac{n_h}{N_h}\right)$$

Population mean

$$\mu = \frac{1}{N} \sum_{h=1}^H \sum_{i=1}^{N_h} x_{hi}$$

Population variance

$$\sigma^2 = \frac{1}{N} \sum_{h=1}^H \sum_{i=1}^{N_h} (x_{hi} - \mu)^2$$

N_h : size of stratum h

n_h : sample size for stratum h

N : total population size

$W_h = \frac{N_h}{N}$: population weight of h

$f_h = \frac{n_h}{N_h}$: sampling fraction in h

$\bar{x}_h$: sample mean in h

s_h^2 : sample variance in h

 The mean and variance are weighted by the stratum's size ($W_h = \frac{N_h}{N}$).

Stratified sampling

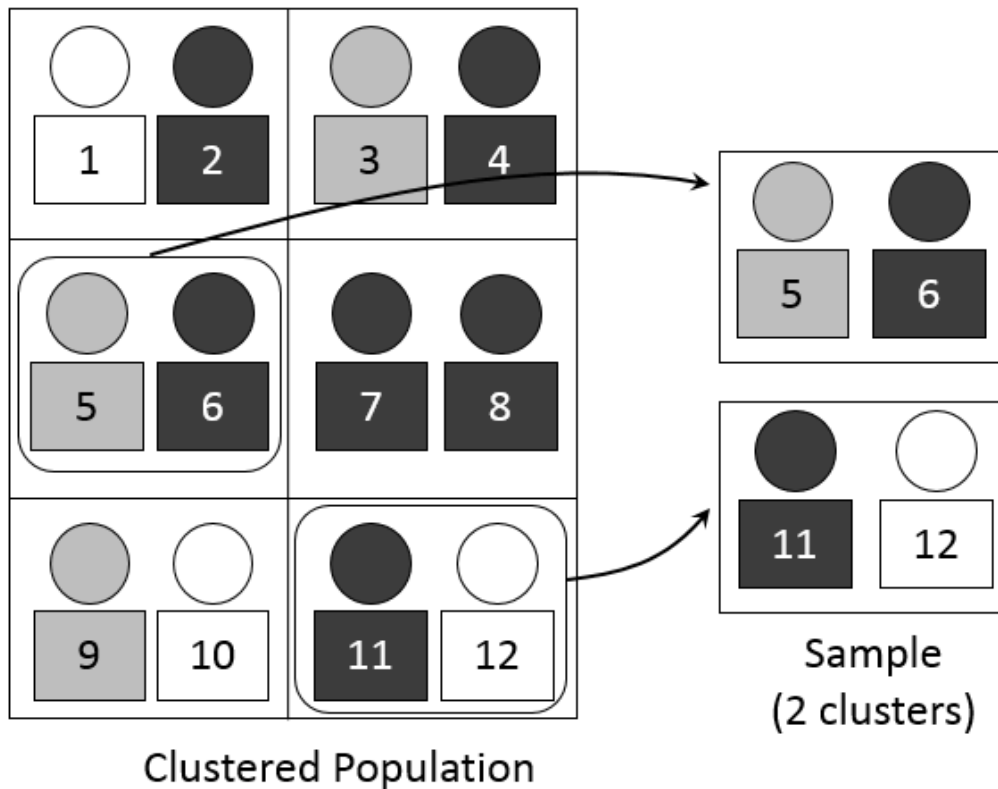
Implementation in R

```
1  set.seed(7)                                # Set seed for reproducibility
2  H <- 3                                       # Number of strata
3  N_h <- c(4000, 3000, 3000)                 # Population sizes for each stratum
4
5  stratum <- rep(1:H, times = N_h)           # Vector labeling each unit's stratum
6  pop <- data.frame(id = 1:sum(N_h),         # Build full population dataset
7                    h = stratum,              #   - stratum membership
8                    y = c(rnorm(N_h[1], 50, 10), #   - outcomes for stratum 1
9                          rnorm(N_h[2], 60, 15), #   - outcomes for stratum 2
10                         rnorm(N_h[3], 55, 12))) #   - outcomes for stratum 3
11
12 N <- nrow(pop)                               # Total population size
13 W_h <- N_h / N                               # Stratum weights (proportion of population)
14 n <- 600                                     # Desired total sample size
15
16 n_h <- round(n * W_h)                       # Allocate sample size per stratum
```

 Try to run the code in your Script Editor.

Cluster sampling

- Population is divided into **clusters**, often based on geography or natural groupings.
- A **random sample of clusters** is selected, and **all (or a sample of) units within selected clusters** are observed.



Cluster sampling

Sample mean

$$\bar{x}_{cl} = \frac{1}{n_c} \sum_{c \in S} \bar{x}_c$$

Approximate variance of the sample mean

$$\text{Var}(\bar{x}_{cl}) = \frac{1}{n_c} \left(1 - \frac{n_c}{C}\right) S_c^2$$

Population mean

$$\mu = \frac{1}{N} \sum_{c=1}^C \sum_{i=1}^{N_c} x_{ci}$$

Population variance

$$\sigma^2 = \frac{1}{N} \sum_{c=1}^C \sum_{i=1}^{N_c} (x_{ci} - \mu)^2$$

N : Population size

C : Number of clusters in the population

N_c : Size of cluster

n_c : Number of clusters sampled

x_{ci} : Value of the i -th element in cluster c

$\bar{x}_c$: Mean of cluster c

$\bar{x}_{cl}$: Mean of sampled clusters

μ : Population mean

σ^2 : Population variance

S_c^2 : Variance of cluster means

S : Set of sampled clusters

 The mean and variance are weighted by the clusters' size similar to stratified sampling ($W_h = \frac{N_h}{N}$) if needed.

Cluster sampling

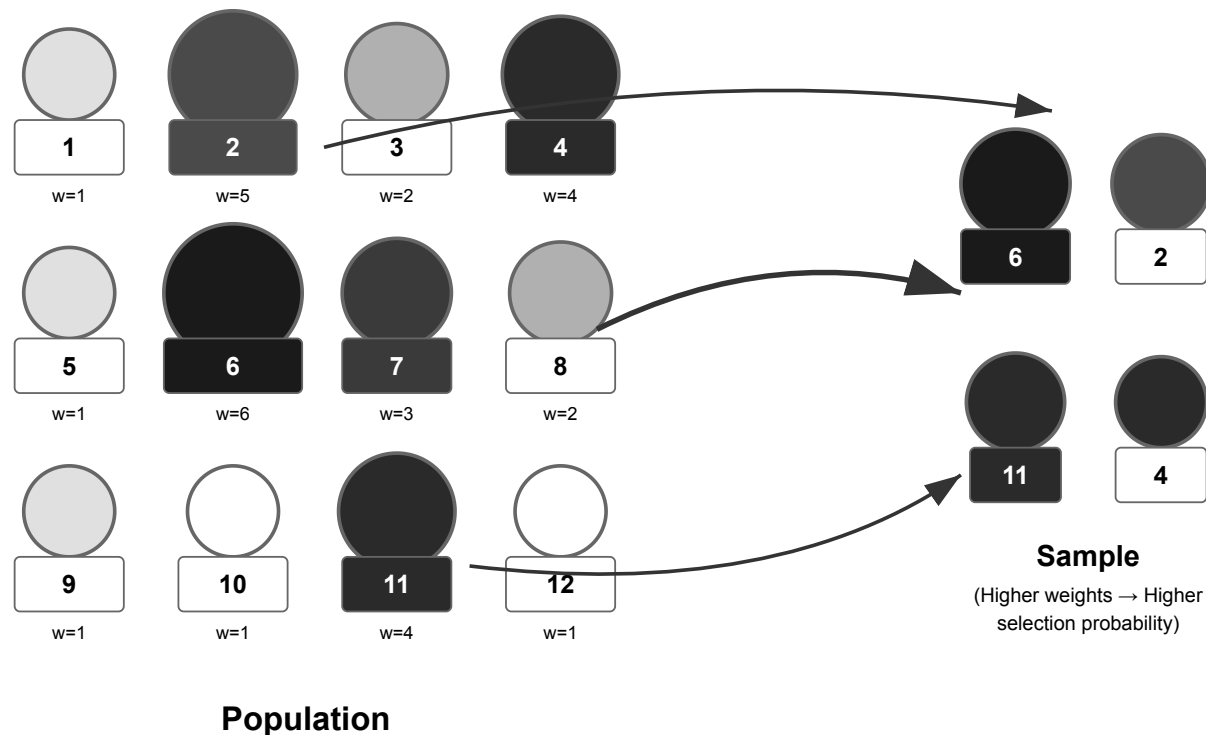
Implementation in R

```
1  set.seed(11)                    # Set random seed for reproducibility
2  J <- 100                        # Number of clusters in the population
3  mbar <- 20                      # Average cluster size
4  rho <- 0.15                     # Intra-cluster correlation (ICC)
5
6  M_j <- rpois(J, mbar-1) + 1     # Cluster sizes (Poisson around mbar)
7  mu <- 50                       # Overall population mean
8  sigma2 <- 100                  # Total variance of outcomes
9  tau2 <- rho * sigma2            # Between-cluster variance component
10 sigma2_within <- (1-rho) * sigma2 # Within-cluster variance component
11 cluster_effect <- rnorm(J, 0, sqrt(tau2)) # Random cluster effects ~ N(0, tau2)
12
13 data <- do.call(rbind, lapply(1:J, function(j){ # Generate population data
14   m <- M_j[j]                                # Size of cluster j
15   y <- mu + cluster_effect[j] +              # Mean + cluster effect +
16     rnorm(m, 0, sqrt(sigma2_within)) # individual error within cluster
```

 Try to run the code in your Script Editor.

Weighted sampling

- Each unit in the population is assigned a **weight** reflecting its **probability of selection**.
- Units with higher weights have **greater influence** on the estimate.
- Often used in **complex surveys** or **unequal probability sampling** designs.



Probability weighted sampling.

Weighted sampling

Weighted sample mean

$$\bar{x}_w = \frac{\sum_{i \in S} w_i x_i}{\sum_{i \in S} w_i}$$

Population total (Horvitz–Thompson estimator)

$$\hat{T} = \sum_{i \in S} \frac{x_i}{\pi_i}$$

Population mean (Horvitz–Thompson estimator)

$$\mu_{ht} = \frac{1}{N} \sum_{i \in S} \frac{x_i}{\pi_i}$$

N : Population size

x_i : Value of the i -th element in the population

w_i : Weight of element i

S : Set of sampled elements

$\bar{x}_w$: Weighted sample mean

μ_{HT} : Population mean

$\hat{T}$: Population total

π_i : Inclusion probability for unit i

w_i : Sampling weight for unit i , $1/\pi_i$

 The Horvitz–Thompson estimator works if N is known.

Weighted sampling

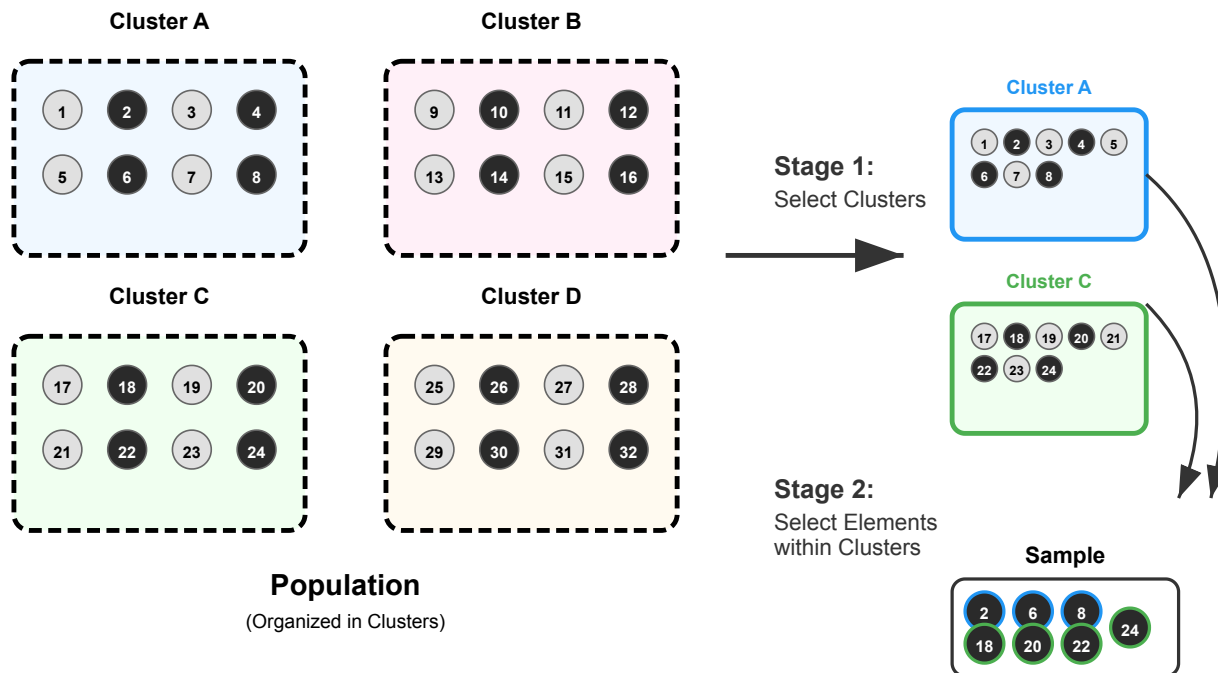
Implementation in R

```
1  set.seed(123)                                # Fix seed for reproducibility
2
3  N <- 1000                                     # Population size
4  x <- runif(N, 1, 10)                         # Size measure
5  y <- 5 + 2*x + rnorm(N)                     # Outcome correlated with x
6  pop <- data.frame(id=1:N, x=x, y=y)         # Store in a dataframe
7
8  n <- 100                                     # Sample size
9  srs <- pop[sample(1:N, n), ]                 # Draw SRS without replacement
10 mean_srs <- mean(srs$y)                     # Sample mean (unweighted)
11 mean(pop$y)                                # True population mean (for comp
12
13 prob <- x / sum(x)                           # Selection probs proportional t
14 samp_ids <- sample(1:N, n, prob=prob)        # Draw weighted sample
15 pps <- pop[samp_ids, ]                      # Extract sampled units
16 pps$w <- 1/prob[samp_ids]                  # Compute weights = 1 / inclusio
```

 Try to run the code in your Script Editor.

Multistage sampling

- Sampling is conducted in **two or more stages**, selecting units at each stage.
- Common design:
 - Stage 1: Randomly sampled **clusters** (e.g., schools).
 - Stage 2: Randomly sampled **elements within clusters** (e.g., students).
- Useful when a complete list of all population units is unavailable.



Multistage sampling

Population mean

$$\mu = \frac{1}{N} \sum_{c=1}^C \sum_{i=1}^{N_c} x_{ci}$$

Population variance

$$\sigma^2 = \frac{1}{N} \sum_{c=1}^C \sum_{i=1}^{N_c} (x_{ci} - \mu)^2$$

2-stage sample mean

$$\bar{x}_{ms} = \frac{1}{n_c} \sum_{c \in S} \bar{x}_c, \quad \text{where } \bar{x}_c = \frac{1}{n_{ci}} \sum_{i \in S_c} x_{ci}$$

Approximate variance of $\bar{x}_{ms}$

$$\text{Var}(\bar{x}_{ms}) \approx \frac{1}{n_c} \left(1 - \frac{n_c}{C}\right) S_c^2 + \frac{1}{n_c^2} \sum_{c \in S} \frac{S_{ci}^2}{n_{ci}}$$

N Total population size

C : Number of clusters in population

N_c : Size of cluster c

n_c : Number of clusters sampled

S : Set of sampled clusters

x_{ci} : Value of element i in c

S_c : Variance of cluster means in population

S_{ci}^2 : Variance of elements within sampled c

n_{ci} : Number of elements sampled from c

$\bar{x}_c$: Mean of sampled elements in c

Multistage sampling

Implementation in R

```
1 set.seed(42) # Ensure reproducibility
2
3 # --- Stage 0: Define population ---
4 J <- 50      # Number of primary clusters (e.g., schools)
5 mbar <- 30   # Average number of units per cluster (e.g., students)
6 mu <- 60    # Overall population mean
7 sigma2 <- 100 # Total variance
8
9 # Generate cluster sizes (first-stage)
10 M_j <- rpois(J, mbar-1) + 1 # Random cluster sizes around mbar
11
12 # Generate cluster-level effects
13 tau2 <- 0.2 * sigma2      # Between-cluster variance
14 sigma2_within <- sigma2 - tau2 # Within-cluster variance
15 cluster_effect <- rnorm(J, 0, sqrt(tau2)) # Random cluster effects
16
```

 Try to run the code in your Script Editor.

Summary table

Sampling Method	Key Characteristics	Strengths	Limitations
Simple Random	Every unit has an equal chance; requires full sampling frame	Unbiased; easy to implement	Inefficient for large or dispersed populations; ignores subgroups
Stratified	Population divided into homogeneous strata; random sample within each stratum	Higher precision; ensures subgroup representation	Requires stratum info for all units; complex analysis
Cluster	Population divided into clusters; some clusters randomly selected	Cost-effective; useful for large/geographic populations	Higher variance if clusters are homogeneous
Weighted	Units sampled with unequal probabilities; weights used in estimation	Adjusts for unequal probabilities; enables oversampling	Requires accurate weights; complex variance estimation
Multistage	Sampling occurs in two or more stages (e.g., clusters and elements)	Logistically efficient; scalable	Can increase variance; complex design and analysis
Non-Probability	Units selected based on convenience or judgment,	Easy, fast, cheap	Cannot generalize to population; prone to

Sampling design

Key considerations

1. Define **population** and **objectives**
2. Establish reliable **sampling frame**
3. Choose **method aligned** with objectives and resources
4. Calculate appropriate **sample size**
5. Minimize **bias and sampling error**
6. Document **design and limitations** thoroughly

Take home

- Sampling allows inference from a subset when a full census is impractical.
- Probability sampling ensures representativeness and enables unbiased estimates.
- Stratification and weighting improve precision and adjust for unequal probabilities.
- Cluster and multistage sampling enhance feasibility for large or dispersed populations.
- Careful design, sample size calculation, and bias minimization are essential for reliable results.

References

Särndal, C.-E., Swensson, B., & Wretman, J. (1992). *Model Assisted Survey Sampling*. Springer.
<https://d-nb.info/969712979/04>

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Lumley, T., & Scott, A. (2023). *survey: Analysis of Complex Survey Samples*. R package version 4.2-3.
<https://cran.r-project.org/web/packages/survey/index.html>

Exercise

Please download the R script with exercises from **GitHub** and try to complete it.